INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Date: 24-02-2026 FN **Time: 2 Hrs** **Full Marks: 30**

Department: Mechanical Engineering

Sub. No.: ME60012 **Mid-Sem Exam** **Computational Fluid Dynamics**

1.) Obtain the second-order accurate forward-difference expression for $\frac{d^3y}{dx^3}$ using uniform grid-spacing. Comment on the values of the errors obtained in evaluating the value of $\frac{d^3y}{dx^3}$ for (i) $y = x^2 + \frac{x}{3}$ and (ii) $y = e^{2-x^2}$ in the domain $0 \le x \le 1$ using this discretization scheme using dx=0.01 in a single precision (8-decimal digit) computer.

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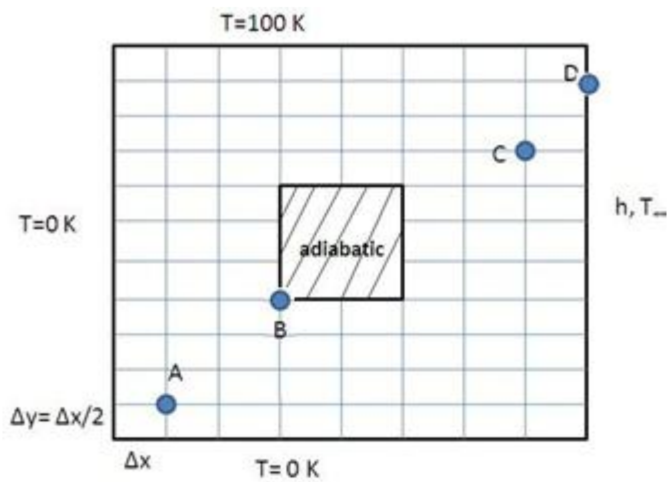
Obtain- $\frac{d^3y}{dx^3} = \frac{-3y_{i+4}+14y_{i+3}-24y_{i+2}+18y_{i+1}-5y_i}{2dx^3}$ (obtaining incomplete expressions with 5 points will award partial marks) ---------------- **2**

For (i) only round-off error- o(10^{-8}) ---------------- **2**

(ii) truncation error – o(0.0001) ---------------- **2**

2.) Figure 1 shows geometry for heat conduction in a 2D grooved plate. Write down the second order finite difference equations for $T(i,j)$ at points A,B,C,D. The expressions should incorporate boundary effects if applicable. **8**



For the point c, general expression:

$$\frac{T_{i-1,j} - 2T_{i,j} + T_{i+1,j}}{(\Delta x)^2} + \frac{T_{i,j-1} - 2T_{i,j} + T_{i,j+1}}{(\Delta y)^2} = 0$$

As $\Delta x = 2\Delta y$, we get $4T_{i,j-1} + T_{i-1,j} - 10T_{i,j} + T_{i+1,j} + 4T_{i,j+1} = 0$ *with* error $O(dx^2)$ ----------- **2**

For point A, j-1 and i-1 are BCs at 0K: $-10T_{i,j} + T_{i+1,j} + 4T_{i,j+1} = 0$ ----------- **2**

For point B, image point to substitute i+1 and j+1 $4T_{i,j-1} + T_{i-1,j} - 5T_{i,j} = 0$ ----------- **2**

For point D, image point to substitute i+1 using convective condition ----------- **2**

3.) Consider one-dimensional unsteady conduction equation $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$ to be solved by Euler scheme in the domain $0 \le x \le 1, t > 0$. Write the discretized equation for any internal point (i) at time-step (n). Consider $\Delta x = 0.01$. Find the largest stable value of time-step, Δt for that if $\alpha = 10^{-3} m^2/sec$. Find the order of error in this case.

Find the largest stable time-step for $\Delta x = 0.1$.

6

For $\Delta x = 0.01$: $\frac{\alpha \Delta t}{x^2} \le \frac{1}{2}$ **(1)** gives $\Delta t_{max} = 0.05$ **(2)**

With error o($\Delta x^2, \Delta t$) - error -o(0.05) **(1)**

For $\Delta x = 0.1$ weget $\Delta t_{max} = 5$ **(1)** error >1 - unstable **(1)**

3.) In a false transient method, a steady problem is considered as an unsteady problem with the exact boundary conditions and an arbitrary initial condition. As the time progresses, the solution converges to the steady state solution, as the boundary conditions are consistent with that.

Now, consider one meter long composite rod whose conductivity depends on temperature as $k(T) = a + bT^{0.3}$. The boundary conditions are given as $T(0) = 0\,; T\,(1) = 1.0$. Define an algorithm to obtain steady state temperature distribution over the rod with an internal heat generation, q per unit length. Also indicate the system of equations (if any) and its solution methodology.

You may assume any other property as constant.

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Unsteady governing equation: $\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial t}\left(k(T)\frac{\partial T}{\partial x}\right) \quad => \rho c_p \frac{\partial T}{\partial t} = \frac{\partial k(T)}{\partial x}\frac{\partial T}{\partial x} + k(T)\frac{\partial^2 T}{\partial x^2}$

Euler FDM expression $\rho c_p \frac{T_i^{(n+1)} - T_i^{(n)}}{\Delta T} = \frac{k_{i+1}^{(n)} - k_{i-1}^{(n)}}{2\Delta x} \times \frac{T_{i+1}^{(n)} - T_{i-1}^{(n)}}{2\Delta x} + k_i^{(n)} \frac{T_{i+1}^{(n)} - 2T_{i-1}^{(n)} + T_{i-1}^{(n)}}{\Delta x^2}$

To solve this in false transient method till steady value comes --------------------- **(5)**

Substitute BCs and arbitrary but physical initial condition --------------------- **(1)**

Matrix formation and multiplication --------------------- **(2)**

Iterations and its convergence of iteration to steady value based on norm/max/rms--------------------- **(1+1)**

[Similarly, marks awarded for full steady solution based on linearlization and subsequent iterations for convergence]